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Speaker Recognition System using Gaussian Mixture Model

Kanyadara Saakshara, Kandula Pranathi, R.M.Gomathi, A. Sivasangari, P. Ajitha and T. Anandhi

Abstract—**The medical field is one of the most important are as that is researched by numerous researchers in order to develop the existing system. Though there have been numerous new and innovative systems developed, there are always some challenges present in the field that needs to be solved. It is important for a healthcare system to make sure that it is secure enough and any leak of the patient data could lead to numerous other risk factors. In this paper, we propose a speaker recognition system that makes use of the Gaussian Mixture Model which identifies a person from characteristics of their voice to authenticate them to access the medical records. The model is able to determine the gender of the person to be authenticated and also recognizes the person. The model uses Cepstral Analysis as the feature extraction technique using the feature Mel Frequency Cepstral Coefficient (MFCC). The proposed work is observed to work well when compared with other existing systems and could be of much use in performing and giving access to the authorized employees in a health care system.**

***Index Terms*—Automated System, Feature extraction, Gaussian Mixture Model, Healthcare, Mel Frequency Cepstral Coefficient Voice Recognition**

I. Introduction

As the medical field deals with the life of a human, immense care needs to be given for the same [1]. It is very essential even to safeguard the information pertaining to the patient details and all other data that is relevant to the healthcare system. In case, any unauthorized person gaining access to the patient details can modify the data and this could result in more harm to the entire health care system [2]. It is very important to know who is accessing the information and who walks in and out of the health care in order to make the entire system more secure [3,4]. All the authorities entering and accessing the system on-line and off-line should be identified and verified first and then given the necessary credentials to have control over the data they are accessing.

The medical field deals with human lives and saving them is the biggest challenge. Many new technologies are used for improving the same. In this paper, we have proposed an automated voice identification or speaker recognition model, that is able to detect the voice of the person and able to recognize the gender and also able to correctly recognize the person. The model is constructed by making use of a classification model. There is no chance of any failure. Our Proposed system was achieved the identification level on comparing with existing system.

The rest of the paper is organized as below. Section II and Section III describes about the related work and proposed approach respectively. The results are discussed in Section IV. At last, Section V concludes the paper with conclusion of the work.

II. Related Work

Different studies are effective and happening with regards to speaker recognition. It plays a crucial part within securing as well as determining someone regarding protection. You will find numerous scientific studies which display many instances of profitable utilization of speech management for movable robots [5], humanoid robots[6],as well as aerial robots [7]. Speech management continues to be achieved growth that is good along with a lot of people as well as during many medical scientists. It's additionally being browsed upon they've quite nominal usage within manufacturing programs. Lately, manufacturing robots are furnished around such a method in which they're made by using a human-to-machine interaction program which in human robot effort is allowed by turn. This is noticed to get better when compared with implementing manpower. [8] Proved all of the attainable advantages which may be gathered up as a result of this particular kind of human robot cohesiveness at work. This was carried out by repairing robotic production and versatility throughput. Because there are plenty of advantages this way, during the current stage of it's of advancement, speech control was awfully challenging being acknowledged by the company [9,10].

Kanyadara Saakshara, Kandula Pranathi, R.M.Gomathi and A. Sivasangari, P. Ajitha and T. Anandhi are with the School of Computing, Sathyabama Institute of Science and Technology, Chennai, India (e-mail: saakshu9876@gmail.com pranathipandu1@gmail.com gomssrm@gmail.com sivasangarikavya@gmail.com hannahgracelyne@gmail.com anandhitamil2908@gmail.com).

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Planned a method which delineated different problems that are inside the business oriented vocal managing methods and also inside the essentials which was needed certain the productive integration of theirs with sector. The validation of robots in attention is in addition comparable which was pertaining towards the company.

Surgical treatment might be a technique of man maintenance; nonetheless, there is contact information of the components' specification. The mixture of golem-assisted MIS and voice management started with laptop computer motion's launch of the entire fabulist automatic robot [11] that was accredited through the federal bureau of 1994. The fabulist golem has a laparoscopic digital camera that is fastened on the robotic arm, around the functioning dining room table. The robotic arm is managed employing a joystick or perhaps by speech instructions. Within time, the primary key down sides of the method had been the latency on the speech recognition motor as well as additionally the lower recognition pace.

KaLAR [12] is a good example of an additional small medical assistant that is furnished by using a seven command vocal structure as well as instrument instruction. Within difference to fabulist, KaLAR is fastened onto the medical dining room table. Throughout 2003, [13], a light weight healthcare instrument golem, which resulted within the development on the ViKY female inner reproductive body organ actuator [14]. This particular golem is placed on the person's belly. The ViKY voice control user interface is triggered by phrases which square degree and then commands. Within an exceptionally relative analysis [15], vocal recognition process accomplished successful number of seventy one, when compared to fabulist process, which contains successful rate of sixty seven. Vocal recognition methods are utilized within several areas as well as one particular area is healthcare.

Highly effective computing devices have been developed by many scientists in [16], has modeled a method which is in a position to understand with approximately 98.9 % of proper identification. Overall it's majorly used-to determine the individual's identity. This model use the real time air quality information in a city by measuring the pollution information using sensors and data sets[17]. In this paper, achieving evidence based drugs analysis is done using big data. This procedure is possible by gathering of medical evidences, grouping of data, Mapping of disease data set and Medicines, and report generation [18]. Data fusion plays a vital role in WSN to saveenergy of the sensors. Wireless Sensor Network has theability to eliminate the redundant data, efficiently giveaccurate data and consequently reduce the energyconsumption. Different Data fusion models and their working principle are studied, analysed and surveyed in order to understand the goals of each data fusion model [19].

III. PROPOSEDAPPROACH

The proposed system can be summarized in five basic phases: Data Acquisition, Data pre- processing, Feature Extraction, Model Training and Performance Testing (Identification). Shown below is the block diagram of the proposed system. From Fig. 1, it can be noted that the first step is the data collection. The sample data is collected from our colleagues, friends and family. Now the input signal must be pre-processed, that is noise reduction and silence removal in order to achieve better outputs and prediction results. In this classification model, the features are extracted from the input signal and are sent further for the data modeling purpose.

The feature extraction is done by making use of two main features: The Mel- Frequency Cepstrum Coefficient (MFCC) and the Delta-MFCC. We calculated 20 MFCC's and 20 Delta-MFCC's and so we had 40 features in hand. MFCC focuses on series of calculations that uses Cepstrum with a nonlinear frequency axis following Mel scale. To obtain melcepstrum, the voice signal is windowed using analysis window and then Discrete Fourier Transform is computed. The main purpose of MFCC is to mimic the behaviour of human ears.

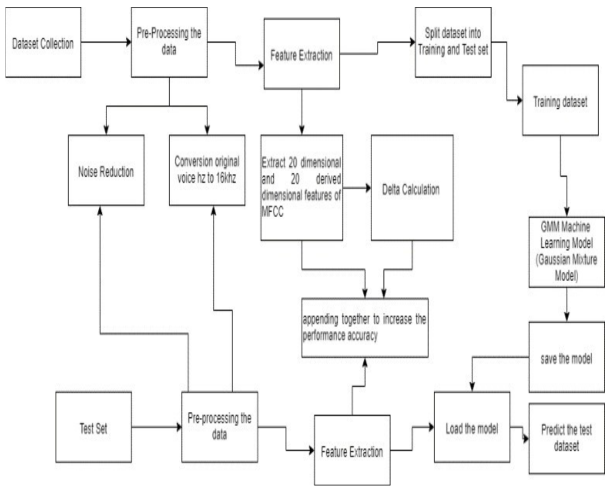

Fig. 1. Block Diagram of the Proposed System

MFCC estimation process is subdivided into six phases or blocks as shown in Fig. 2. We implemented the Gaussian Mixture Model (GMM) approach for model training. The GMM is a probabilistic model that assumes all the data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters. We can assume mixture models as generalizing k-means clustering to incorporate information about the covariance structure of the data as well as the centers of the latent Gaussians as shown in Fig. 3. The speaker identification system can be separated into four modules: Front-end processing, Speaker modeling, Speaker database and Decision logic.

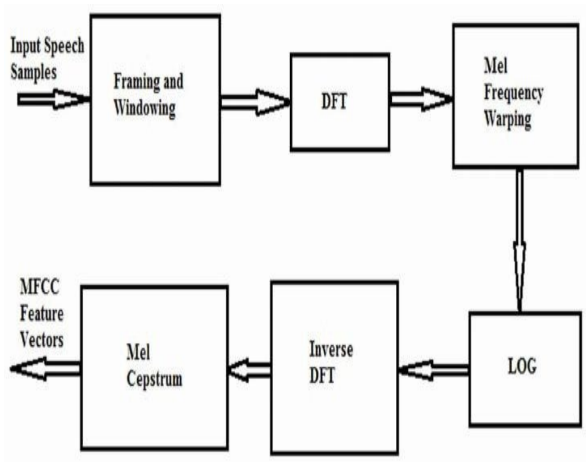

Fig. 2. Phases of MFCC estimation

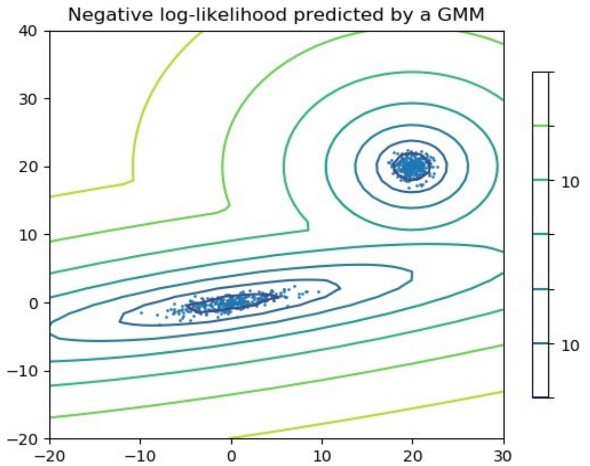

Fig. 3. Two-component Gaussian mixture model: data points and equi-probability surfaces of the model

We can infer from Fig. 5 that the audio sample given is tested and the speaker of the audio sample is identified with maximum accuracy.

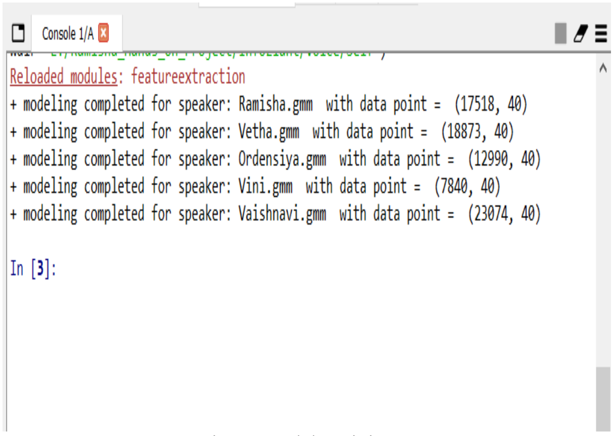

Fig. 4. Model Training

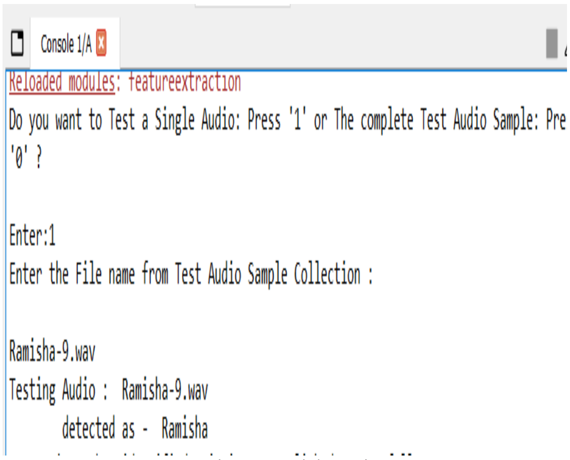

Fig. 5. Model Testing

IV. Experimental Results

The experimental results were done on various datasets. We tested our model's working and accuracy on a download dataset that is the VoxForge dataset and also the self-made dataset. The voice recognition is one of the critical aspects that need to be trained upon for maintaining the security of a system. The system was able to efficiently detect the individual along with their gender by making use of the tone, voice and speech patterns. The model was able to recognize who was speaking with automatic feature extraction. Fig. 4 shows the training of the GMM with the 40 features extracted using MFCC and Delta- MFCC.

V. Conclusion

As one in every maximum advancing field in era for treating patients, it's also critical that the medical establishments must have a secured data preservation system so that only authenticated individuals may have access to them. One way to authenticate is to recognize and identify the individuals that are accessing the records. The proposed speaker recognition was successfully conducted with an outstanding result on the self made dataset with an accuracy of 95.29%. The proposed MFCC-GMM model could be claimed to perform better than the SVM classifier as it produced only 83.33%.The proposed model has efficiently produced the results and various parameters are also used to measure the performance of the system.

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